

MEININGEN, Germany

WMO: 105480

Lat: 50.5611N

Lon: 10.3772E

Elev: 1486

StdP: 13.92

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.8	11.6	-1.8	5.3	11.0	3.5	7.0	12.3	27.1	39.8	23.8	37.6	5.1	20	0.593

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	80.9	63.6	77.8	62.5	74.6	61.1	65.9	76.9	63.9	74.0	62.3	71.9	7.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.0	87.7	69.8	60.1	82.0	68.0	58.4	77.0	66.7	31.4	77.0	29.9	73.8	28.7	72.0	71.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.1	18.0	15.8	4.1	87.6	6.9	2.6	-0.9	89.5	-4.9	91.0	-8.8	92.4	-13.8	94.3		
(5)			3.1	69.5	7.0	1.9	-1.9	70.8	-6.0	72.0	-9.9	73.0	-15.0	74.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.1	29.2	28.8	36.8	43.1	52.2	57.3	61.6	61.4	54.8	46.4	36.5	32.2
	DBStd	13.62	9.11	8.20	6.92	6.54	7.02	6.57	6.03	5.84	5.57	6.34	6.47	7.26
	HDD50	3059	646	593	411	222	61	9	0	0	15	146	405	551
	HDD65	7342	1111	1013	875	658	399	245	143	143	307	577	855	1016
	CDD50	1277	0	0	1	15	128	227	359	354	159	34	0	0
	CDD65	83	0	0	0	0	1	13	36	31	2	0	0	0
	CDH74	793	0	0	0	0	16	117	331	303	26	0	0	0
	CDH80	128	0	0	0	0	0	14	52	62	0	0	0	0
Wind	WSAvg	6.4	7.7	6.8	7.1	6.4	5.8	5.8	5.6	5.6	6.2	6.4	6.9	6.8
Precipitation	PrecAvg	25.6	2.2	1.7	1.9	1.5	2.2	2.6	3.0	2.4	2.2	2.0	2.2	2.4
	PrecMax	31.7	4.0	4.6	4.2	3.1	6.3	5.7	5.9	5.5	4.9	6.6	3.9	4.6
	PrecMin	19.3	0.6	0.2	0.4	0.1	0.1	0.5	0.9	0.4	0.4	0.3	0.0	0.4
	PrecStd	3.2	0.9	1.0	1.0	0.7	1.3	1.3	1.3	1.4	1.0	1.2	0.9	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.4	52.2	61.2	67.0	76.4	82.3	84.8	86.5	77.6	67.5	52.8	51.1
		MCWB	45.2	47.1	49.4	53.0	59.4	65.5	66.9	65.6	59.6	57.2	48.8	47.4
	2%	DB	44.6	46.3	54.5	62.8	72.0	77.4	81.0	80.8	73.3	62.5	50.2	47.3
		MCWB	42.1	43.3	46.5	50.8	57.1	62.9	64.0	62.9	60.8	54.9	47.3	44.7
	5%	DB	42.3	42.1	50.1	59.1	68.9	73.7	77.9	77.2	69.0	58.6	47.6	44.8
		MCWB	40.0	39.4	43.9	48.9	56.1	60.9	63.1	62.0	58.5	53.2	45.2	42.6
	10%	DB	40.1	38.9	46.7	55.3	65.6	69.9	74.3	73.5	65.3	55.6	45.2	42.5
		MCWB	38.0	36.5	42.1	46.5	54.6	59.0	61.6	60.6	57.0	51.4	42.7	40.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.3	47.6	50.8	54.4	61.7	67.6	68.9	68.8	63.8	59.2	50.2	47.9
		MCDB	47.6	51.5	58.0	64.3	71.1	78.7	81.8	82.2	70.6	65.2	51.9	49.7
	2%	WB	42.5	43.4	47.3	51.8	59.0	64.2	66.3	65.1	61.3	56.0	47.6	45.0
		MCDB	44.3	46.2	53.3	60.8	69.2	73.5	77.3	76.5	69.6	60.2	49.6	46.8
	5%	WB	40.4	39.8	44.9	49.4	57.0	62.1	64.2	63.3	59.6	54.0	45.3	42.8
		MCDB	42.1	41.7	49.7	57.5	67.2	71.0	74.6	73.8	67.2	58.0	47.4	44.7
	10%	WB	38.1	36.8	42.4	47.4	55.1	60.0	62.4	61.7	57.9	51.9	43.0	40.7
		MCDB	39.9	38.8	46.2	54.4	64.4	68.4	72.0	71.1	64.5	55.4	44.8	42.3
Mean Daily Temperature Range	5% DB	MDBR	7.7	10.2	12.0	15.2	16.5	15.2	17.1	17.5	15.2	12.5	8.4	7.1
		MCDBR	8.8	13.4	18.6	22.1	21.9	21.8	23.8	24.3	22.0	17.2	10.8	9.1
		MCWBR	8.1	10.7	12.4	13.1	11.8	11.0	11.1	10.7	11.5	10.8	9.2	8.3
	5% WB	MCDBR	8.6	13.1	17.1	20.2	20.4	20.0	21.3	20.9	18.8	15.2	10.2	8.8
		MCWBR	8.2	11.0	12.1	12.3	11.8	10.9	11.0	10.4	10.8	10.2	9.0	8.3
Clear-Sky Solar Irradiance	taub	0.298	0.320	0.356	0.393	0.398	0.397	0.409	0.402	0.370	0.346	0.323	0.295	
	taud	2.415	2.370	2.322	2.258	2.273	2.294	2.260	2.302	2.376	2.437	2.439	2.413	
	Ebn at Noon	235	258	267	269	272	273	267	263	260	245	222	217	
	Edh at Noon	19	26	33	39	40	40	41	37	31	24	19	17	
All-Sky Solar Radiation	RadAvg	245	475	816	1292	1535	1679	1592	1373	986	577	270	184	
	RadStd	43	72	109	167	179	144	191	140	139	71	38	22	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+90	N/A	

Nomenclature: See separate page